

PRIMARY 2 MATHEMATICS SCHEME OF WORK

FIRST TERM

Week	Topic	Breakdown (Subtopics)
1	Whole Numbers (1–100)	• Revision of Primary 1 work • Counting numbers 1–100 • Reading and writing numbers 1–100 in figures and words • Comparing numbers using place values (tens and units) • Ordering numbers from smallest to largest and vice versa • Skip counting in 2s, 3s, 5s, and 10s • Identifying odd and even numbers
2	Whole Numbers (101–200)	• Counting, reading, and writing numbers 101–200 • Identifying place values (hundreds, tens, and units) • Writing expanded forms of numbers • Comparing numbers using $<$, $>$, $=$ symbols • Skip counting in 2s, 3s, and 5s up to 200
3	Fractions (Part 1)	• Meaning of fractions • Identifying $\frac{1}{2}$ and $\frac{1}{4}$ as parts of a whole • Dividing objects into two and four equal parts • Writing simple fractions ($\frac{1}{2}$, $\frac{1}{4}$) • Representing fractions with objects and pictures
4	Fractions (Part 2)	• Identifying $\frac{3}{4}$ of given objects • Practical activities on sharing items into four equal parts • Relating fractions to real-life situations (food,

		toys, etc.) • Comparing simple fractions ($\frac{1}{2}$ and $\frac{1}{4}$) • Word problems involving halves and quarters
5	MID-TERM EXAMINATION	• Assessment of Weeks 1–4
6	MID-TERM BREAK	• Holiday / Rest
7	Addition (Part 1)	• Meaning of addition • Identifying the addition sign (+) • Adding single-digit numbers (without carrying) • Addition using number lines and real objects • Word problems on addition • Real-life applications: total number of items
8	Addition (Part 2)	• Addition of 2-digit numbers (with and without carrying) • Addition of three or more numbers • Mental addition of small numbers • Word problems on addition • Checking addition using subtraction
9	Subtraction (Part 1)	• Meaning of subtraction • Identifying subtraction sign (–) • Subtraction as “take away” and “difference between” • Subtracting single-digit and 2-digit numbers without borrowing • Real-life subtraction problems (items remaining)
10	Subtraction (Part 2)	• Subtracting 2-digit numbers with borrowing • Checking subtraction using addition • Word problems on subtraction • Real-life activities involving subtraction (money, distance, items)
11	REVISION	• Review of: Whole numbers, fractions,

		addition, and subtraction
12	EXAMINATION	• End of Term Assessment
13	CLOSING/VACATION	• Term ends



PRIMARY 2 MATHEMATICS LESSON NOTES

FIRST TERM

WEEK 1: WHOLE NUMBERS (1--100).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Count, read, and write numbers from 1 to 100 in figures and words.
- Identify the place value of numbers (Tens and Units).
- Order numbers from smallest to largest and vice versa.
- Identify odd and even numbers and skip count in 2s, 5s, and 10s.

CONTENT



Counting, Reading, and Writing Numbers (1--100): Numbers help us count things. We write numbers in figures (like 1, 2, 3) and in words (like one, two, three). It is like having a secret code where every number has a name.

- **1 to 10:** One, Two, Three, Four, Five, Six, Seven, Eight, Nine, Ten.

- **11 to 20:** Eleven, Twelve, Thirteen, Fourteen, Fifteen, Sixteen, Seventeen, Eighteen, Nineteen, Twenty.
- **Tens Pattern:** After twenty, we use the tens words: Thirty (30), Forty (40), Fifty (50), Sixty (60), Seventy (70), Eighty (80), Ninety (90), One hundred (100).

Examples:

- **45** is written as *Forty-five* ($40 + 5$).
- **100** is written as *One hundred*. It is our biggest number this week.

Place Value (Tens and Units): Every number has a place, like a house for digits. For numbers up to 99, we have two houses: the **Tens** house and the **Units** house.

- The last number on the right is in the **Units** house. Units are single items (1 to 9).
- The number before it is in the **Tens** house. Tens are groups of ten.

Example: 24 Tens | Units 2 | 4

- The 4 is in the **Units** place. It means 4 single ones.
- The 2 is in the **Tens** place. It means 2 groups of ten, which is 20.
- So, $24 = 2 \text{ Tens and } 4 \text{ Units} = 20 + 4$.

Ordering Numbers: We can arrange numbers in two ways, like lining up from shortest to tallest.

- **Ascending Order:** From smallest to biggest (going up).
- *Example:* 5, 10, 15, 20. Think of walking up stairs, each step is a bigger number.
- **Descending Order:** From biggest to smallest (going down).
- *Example:* 20, 15, 10, 5. Think of walking down stairs.

Odd and Even Numbers and Skip Counting:

- **Even Numbers:** Numbers that make perfect pairs. If you have an even number of shoes, every shoe has a partner. They end in 0, 2, 4, 6, or 8. (e.g., 2, 4, 10, 22).
- *Example:* 6 shoes can be paired as () () () - No shoe is alone.

- **Odd Numbers:** Numbers that have one left out. They end in 1, 3, 5, 7, or 9. (e.g., 1, 3, 11, 25).
- *Example:* 7 shoes: () () ()  - One shoe is alone.

Skip Counting: This means jumping over numbers while counting, like hopping on a number line.

- **In 2s:** 2, 4, 6, 8, 10... (Like counting pairs of legs: 2 legs for one person, 4 legs for two people).
- **In 5s:** 5, 10, 15, 20... (Like counting fingers on one hand, then two hands).
- **In 10s:** 10, 20, 30, 40... (Like counting toes in a group of people).

EVALUATION

1. Write 56 in words.
2. Write "Seventy-two" in figures.
3. In the number 83, what is the place value of 8?
4. Arrange these numbers from smallest to biggest: 12, 5, 30, 20.
5. Is the number 11 an Odd or Even number?
6. Complete the pattern: 2, 4, 6, __, __. (Hint: Skip counting in 2s)
7. How many Tens are in the number 90?

ASSIGNMENT

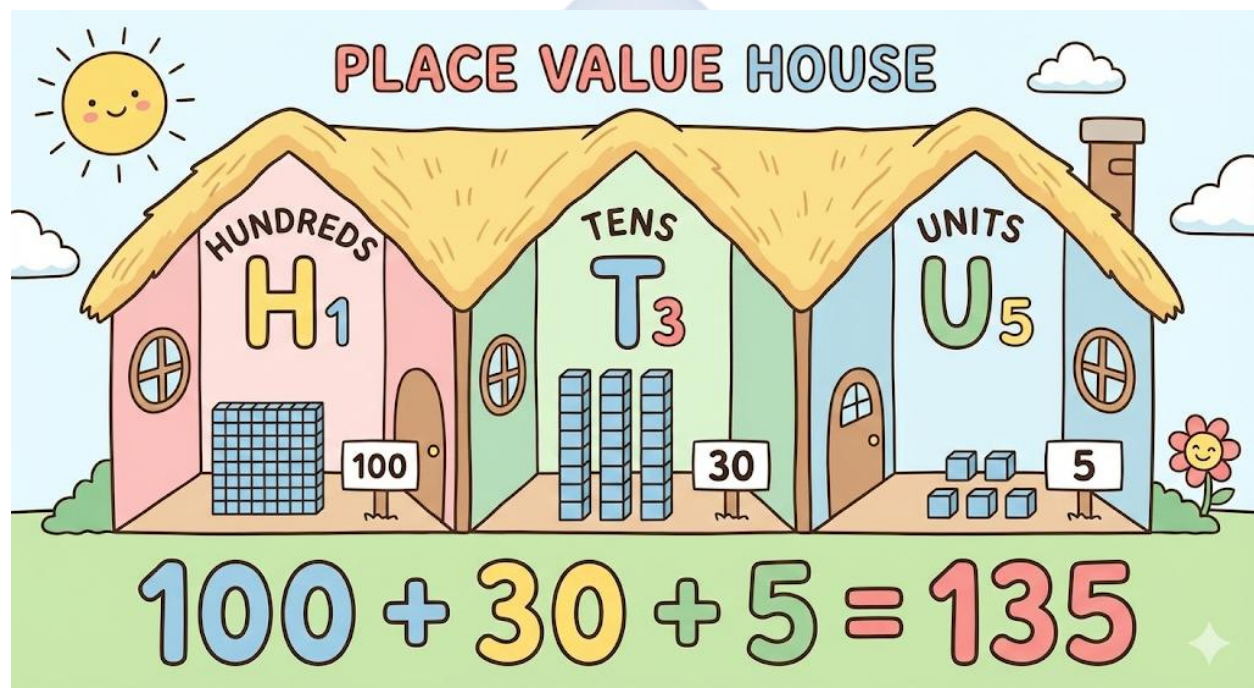
1. Write the numbers 1 to 20 in words.
 2. Write down all the Even numbers between 1 and 10.
 3. Draw a table showing Tens and Units for these numbers: 14, 25, 67.
-

WEEK 2: WHOLE NUMBERS (101--200).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Count, read, and write numbers from 101 to 200.
- Identify place values up to Hundreds, Tens, and Units.
- Write numbers in expanded form.
- Compare numbers using the symbols $<$ (less than), $>$ (greater than), and $=$ (equal to).

CONTENT



Counting and Writing Numbers (101--200): After 100, we start a new hundred. We count by adding numbers to 'one hundred'.

- **101** = One hundred and one. $(100 + 1)$
- **110** = One hundred and ten. $(100 + 10)$
- **150** = One hundred and fifty. $(100 + 50)$
- **200** = Two hundred. $(100 + 100)$
- Think of it like a train: Engine (100) + more carriages (the other numbers).

Place Value (Hundreds, Tens, Units): When we have three digits, we add a new house called **Hundreds**. Now we have three houses from left to right: Hundreds (H), Tens (T), Units (U).

Example: 135 H | T | U 1 | 3 | 5

- 1 is in the **Hundreds** place: It means 1 group of one hundred = 100.
- 3 is in the **Tens** place: It means 3 groups of ten = 30.
- 5 is in the **Units** place: It means 5 single ones = 5.
- So, $135 = 100 + 30 + 5$.

Expanded Form: This means stretching the number out to show the value of each digit. It is like unpacking a bag to see what is inside.

- $124 = 1 \text{ Hundred} + 2 \text{ Tens} + 4 \text{ Units}$.
- *Expanded form:* $100 + 20 + 4$.
- $150 = 1 \text{ Hundred} + 5 \text{ Tens} + 0 \text{ Units}$.
- *Expanded form:* $100 + 50 + 0$ or just $100 + 50$.

Comparing Numbers (<, >, =): We use symbols to show which number is bigger or if they are the same. Think of the symbol as a hungry crocodile's mouth—it always wants to eat the bigger number!

- **> (Greater than):** The open mouth faces the bigger number.
- *Example:* $150 > 100$. The crocodile eats 150 because it is bigger.
- $150 > 100 \Rightarrow$
- **< (Less than):** The pointed end faces the smaller number.
- *Example:* $90 < 120$. The crocodile eats 120.
- $90 < 120 \Rightarrow$
- **= (Equal to):** Used when numbers are the same. Like a balance scale.

- *Example:* $200 = 200$.

EVALUATION

1. Write 145 in words.
2. Write "Two hundred" in figures.
3. What is the place value of 1 in the number 167? (Is it Hundreds, Tens, or Units?)
4. Expand the number 182. (Write it as $100 + \dots$)
5. Put the correct sign ($<$ or $>$): $150 \underline{\hspace{1cm}} 105$.
6. Put the correct sign ($<$ or $>$): $199 \underline{\hspace{1cm}} 200$.
7. Which is bigger: 134 or 143?

ASSIGNMENT

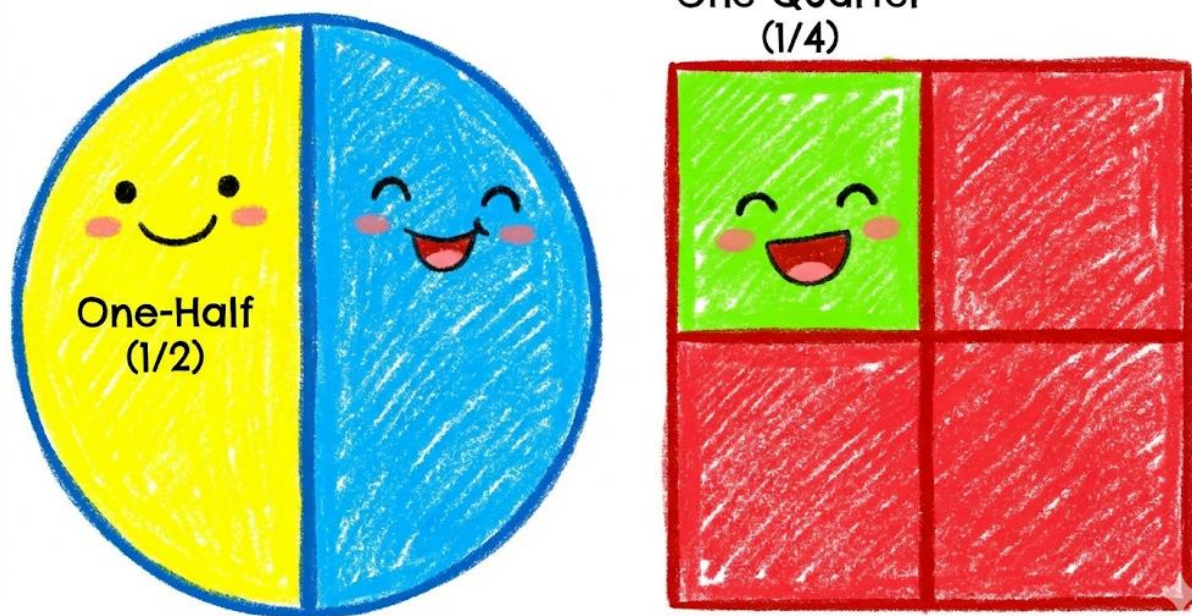
1. Write these numbers in expanded form: 112, 176, 190.
 2. Fill in the correct symbol ($<$, $>$, $=$): a) $120 \underline{\hspace{1cm}} 120$ b) $101 \underline{\hspace{1cm}} 110$
 3. Write the number for: One hundred and sixty-eight.
-

WEEK 3: FRACTIONS (PART 1).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define a fraction as a part of a whole.
- Identify and write the fraction $\frac{1}{2}$ (one-half).
- Identify and write the fraction $\frac{1}{4}$ (one-quarter).
- Represent fractions using objects and drawings.

CONTENT



Meaning of Fractions: A fraction is a part of a whole object. If you have one whole orange and you cut it into pieces, each piece is a **fraction** of the orange. The word 'fraction' means a broken piece.

2. One-Half ($\frac{1}{2}$): When we divide a whole object into **two equal parts**, each part is called **one-half**.

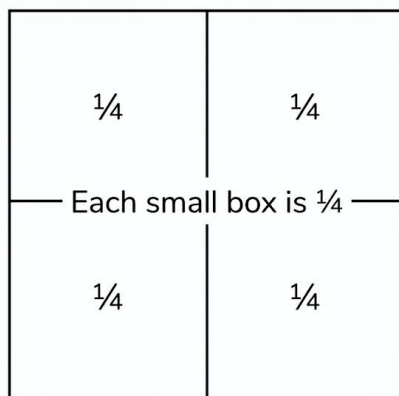
- We write it as $\frac{1}{2}$. The top number (1) is the part we have.
- The bottom number (2) is the total number of equal parts.
- *Example:* Cutting an apple into two equal pieces for two friends. Each friend gets $\frac{1}{2}$.

- Whole Apple: 🍏
- Cut in Half: 🍏 ($\frac{1}{2}$) and 🍏 ($\frac{1}{2}$)

3. One-Quarter ($\frac{1}{4}$): When we divide a whole object into **four equal parts**, each part is called **one-quarter**.

- We write it as $\frac{1}{4}$.
- *Example:* Cutting a square paper into four equal squares. Each small square is $\frac{1}{4}$.

Whole Square:



Representing Fractions: We can draw shapes to show fractions. The most important rule is that the parts must be **equal** in size.

- **To show $\frac{1}{2}$:** Draw a circle and cut it into two equal halves. Shade one half. ●●● ●●●
●●●●● ●●●●● ●●●●● ●●●●● (Shade the left side)
- **To show $\frac{1}{4}$:** Draw a square, divide it into four equal smaller squares, and shade one part.

EVALUATION

1. What is a fraction?
2. How do we write "one-half" in figures?
3. How many halves make one whole? (If you put two halves together, what do you get?)
4. How do we write "one-quarter" in figures?
5. If I cut a loaf of bread into 4 equal parts, what fraction is one part?
6. Draw a circle and shade $\frac{1}{2}$ of it.

7. Draw a rectangle and shade $\frac{1}{4}$ of it.

ASSIGNMENT

1. Draw a square and divide it into 2 equal parts. Shade one part. Write the fraction.
 2. Draw a circle and divide it into 4 equal parts. Shade one part. Write the fraction.
 3. Which is bigger: a whole apple or half an apple?
-



WEEK 4: FRACTIONS (PART 2).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Identify and write the fraction $\frac{3}{4}$ (three-quarters).
- Compare the sizes of $\frac{1}{2}$ and $\frac{1}{4}$.
- Share items into four equal parts practically.
- Solve simple word problems involving halves and quarters.

CONTENT



Three-Quarters ($\frac{3}{4}$): If we divide an object into 4 equal parts and we take 3 of those parts, we have **three-quarters**.

- We write it as $\frac{3}{4}$. It means 3 out of 4 equal parts.
- *Example:* A chocolate bar divided into 4 pieces. If you eat 3 pieces, you ate $\frac{3}{4}$ of the chocolate.
- Chocolate: ☐☐☐☐ (4 pieces)
- You eat: ☒☒☒☐ (3 pieces eaten = $\frac{3}{4}$)

2. Comparing $\frac{1}{2}$ and $\frac{1}{4}$: Let's compare using a circle cut into pieces.

- **Half ($\frac{1}{2}$):** (Big piece)
- **Quarter ($\frac{1}{4}$):** (Small piece)

The $\frac{1}{2}$ piece is bigger than the $\frac{1}{4}$ piece. Sharing with 2 people gives you a bigger piece than sharing with 4 people.

3. Practical Sharing:

- **Finding $\frac{1}{4}$:** If we have 4 sweets and we want to find $\frac{1}{4}$, we share them into 4 equal groups. Each group gets 1 sweet.
- Sweets: 
- Groups: () () () () Each = $\frac{1}{4}$
- **Finding $\frac{1}{2}$:** If we have 8 pencils and want to find $\frac{1}{2}$, we share them into 2 equal groups. Each group gets 4 pencils.
- Pencils: 
- Half: (Group 1 = 4) (Group 2 = 4)

4. Word Problems:

- **Problem:** Mummy has an orange. She gives half to Tolu. What fraction does Tolu have?
- *Answer:* $\frac{1}{2}$. (One out of two equal parts).
- **Problem:** Daddy cut a cake into 4 equal parts. He ate one part. What fraction did he eat?
- *Answer:* $\frac{1}{4}$. (One out of four equal parts).

EVALUATION

1. Write the fraction for "three-quarters".
2. Which is bigger: $\frac{1}{2}$ or $\frac{1}{4}$?
3. If I have 3 parts out of 4, what is the fraction?

4. Divide 4 books into 4 equal groups. How many books are in one group? What fraction is that?
5. Shade $\frac{3}{4}$ of a square. (Hint: Divide into 4, shade 3 parts).
6. How many quarters make one whole? (If you have 4 quarters, what can you make?)

ASSIGNMENT

1. Draw a circle and shade $\frac{3}{4}$ of it.
 2. If you share 8 oranges into 4 equal parts, how many oranges will be in one part? What fraction is that?
 3. Write the fractions: One-half, One-quarter, Three-quarters.
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WEEK 5 & 6: MID-TERM EXAMINATION AND MID-TERM BREAK.

(Note: No lesson notes are required for these weeks as they are for assessment and holiday.)

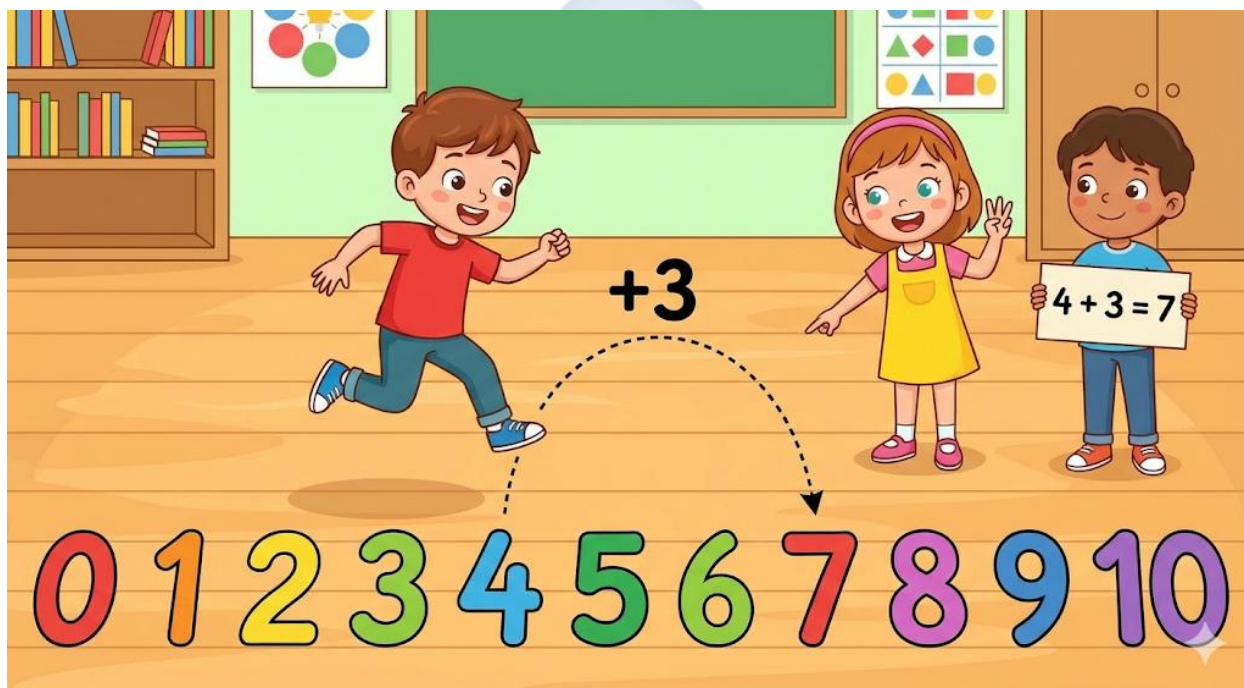


WEEK 7: ADDITION (PART 1).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define addition as putting things together.
- Identify the addition sign (+).
- Add single-digit numbers correctly.
- Solve simple addition word problems using real objects or number lines.

CONTENT



Meaning of Addition: Addition means bringing two or more groups of objects together to make one big group. It means "**finding the total**" or "**how many altogether.**" Think of it as gathering your toys into one box.

2. The Addition Sign (+): The sign for addition is a cross: +.

- It is read as "**plus**" or "**add.**"
- The answer is called the "**sum**" or "**total.**"
- *Example:* $3 + 2 = 5$. (Three *plus* two *equals* five).

3. Adding Single-Digit Numbers:

- $3 + 2 = ?$
- Count 3 fingers. Count 2 more fingers. Count them all together: 1, 2, 3, 4, 5.
- So, $3 + 2 = 5$.
- $4 + 0 = 4$ (Adding zero changes nothing. It's like having 4 apples and getting no more apples).

4. Using Number Lines: A number line helps us jump forward to add. To do $5 + 3$: Start at 5 on the line. Jump forward 3 steps.

0---1---2---3---4---5---6---7---8---9---10 Start → (-1-) (-2-) (-3-)

- Jump 1: from 5 to 6.
- Jump 2: from 6 to 7.
- Jump 3: from 7 to 8.
- The answer is 8.

5. Word Problems:

- Ada has 2 apples. Jo has 3 apples. How many apples do they have altogether?
- We put them together: $2 + 3$.
- $2 + 3 = 5$ apples.

EVALUATION

1. What is the sign for addition?
2. What is $4 + 3$?
3. What is $6 + 0$?
4. Add 5 and 4. ($5 + 4 = ?$)
5. Use a number line to find $2 + 5$. (Draw a simple line from 0 to 10).
6. If I have 1 book and you have 6 books, how many books total?
7. $8 + 1 = ?$

ASSIGNMENT

1. Solve: $3 + 3 = ?$
 2. Solve: $7 + 2 = ?$
 3. Bola has 4 pencils. She buys 1 more. How many pencils does she have now? Write the number sentence.
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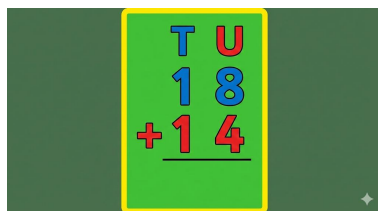


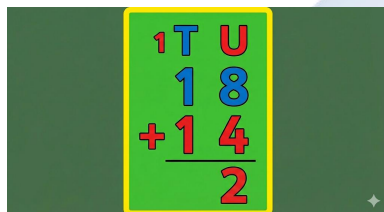
WEEK 8: ADDITION (PART 2).

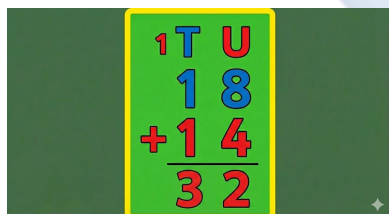
OBJECTIVES By the end of the lesson, pupils should be able to:

- Add 2-digit numbers without carrying.
- Add 2-digit numbers with carrying (renaming).
- Perform mental addition of small numbers.
- Check addition answers using subtraction.

CONTENT


$$\begin{array}{r} \text{T} \quad \text{U} \\ 18 \\ + 14 \\ \hline \end{array}$$


$$\begin{array}{r} \text{T} \quad \text{U} \\ 18 \\ + 14 \\ \hline 2 \end{array}$$


$$\begin{array}{r} \text{T} \quad \text{U} \\ 18 \\ + 14 \\ \hline 32 \end{array}$$

Addition of 2-Digit Numbers (No Carrying): We arrange the numbers vertically under their place value columns: Tens (T) and Units (U). We always add the Units first, then the Tens.

- *Example:* $24 + 13$ T U 2 4
- $13 \text{ ————— } 37$
- **Step 1 (Units):** $4 + 3 = 7$. Write 7 under Units.
- **Step 2 (Tens):** $2 + 1 = 3$. Write 3 under Tens.

- *Answer: 37.*

Addition with Carrying (Renaming): Sometimes the Units add up to 10 or more. Since the Units column can only hold one digit (0-9), we keep the unit part and **carry** the ten part to the Tens column.

- *Example:* $18 + 14$ 1 <-- Carry 1 Ten T U 1 8
- $14 \text{ ————— } 32$
- **Step 1 (Units):** $8 + 4 = 12$. (12 is 1 Ten and 2 Units).
- Write down **2** under the Units column.
- **Carry** the **1** (Ten) to the top of the Tens column.
- **Step 2 (Tens):** Add the carried 1 to the other Tens: $1 \text{ (carried)} + 1 + 1 = 3$.
- Write **3** under the Tens column.
- *Answer: 32.*

Mental Addition: Using our heads for small numbers without writing.

- Think of $10 + 5$. Say "Ten", then count on 5 more: 11, 12, 13, 14, 15. So, $10 + 5 = 15$.
- *Tips:* For $9 + 6$, think " $9 + 1 = 10$ ", then " $10 + 5 = 15$ ". So, $9 + 6 = 15$.

Checking Addition: We can check if our answer is correct by using subtraction, which is the opposite of addition.

- If $2 + 3 = 5$, then $5 - 3$ should equal 2.
- *Check:* $18 + 14 = 32$. Is it true? $32 - 14$ should equal 18.

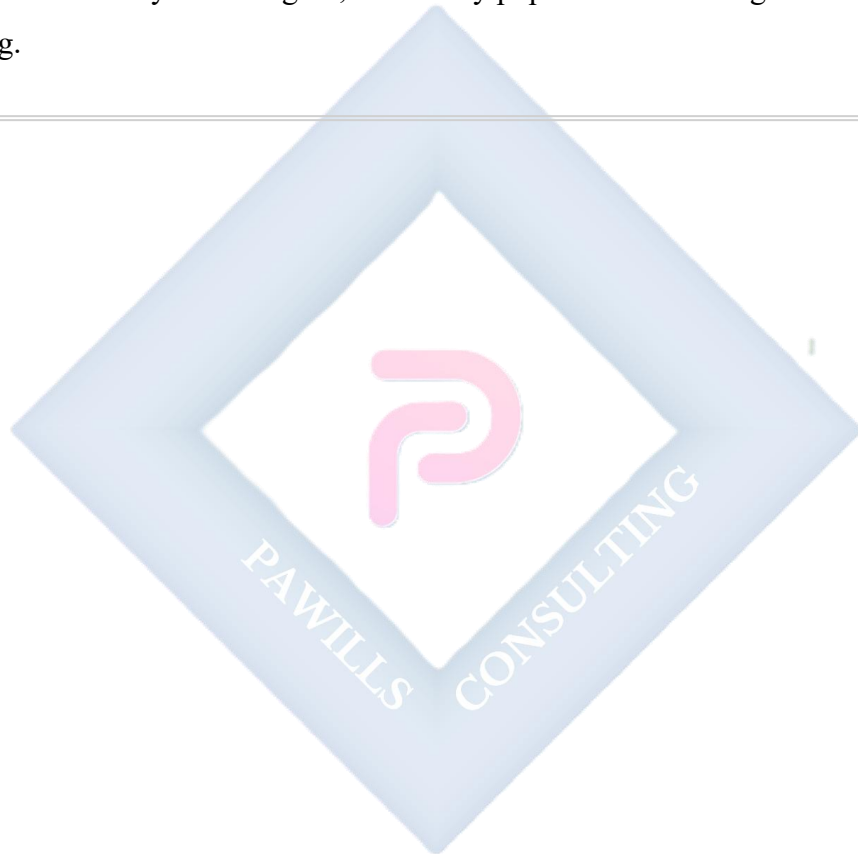
EVALUATION

1. Add $22 + 10$. (Set it out in columns).
2. Add $45 + 13$.
3. Solve: $19 + 11$ (Show your working with carrying).

4. Solve: $27 + 5$. (Hint: $7+5=12$, remember to carry).
5. Calculate mentally: $10 + 10$.
6. Add 3 numbers: $2 + 3 + 4$.

ASSIGNMENT

1. Set out and solve: $36 + 21$.
 2. Set out and solve: $28 + 14$ (Remember to carry).
 3. If a class has 15 boys and 15 girls, how many pupils are there altogether? Show your working.
-



WEEK 9: SUBTRACTION (PART 1).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define subtraction as taking away.
- Identify the subtraction sign ($-$).
- Subtract single-digit numbers.
- Subtract 2-digit numbers without borrowing.

CONTENT



Meaning of Subtraction: Subtraction means taking something away from a group or finding the **difference** between two numbers.

- *Keywords:* Take away, Minus, Less, Remainder, Left over.

2. The Subtraction Sign ($-$): The sign is a short dash: $-$.

- *Example:* $5 - 2$ reads "**5 minus 2**" or "**5 take away 2**".

3. Subtracting Single Digits:

- **$5 - 2$:** Hold up 5 fingers. Fold down 2 fingers. How many are left standing? 3.

- **Number Line:** Start at 5, jump **backwards** 2 times.
- 0---1---2---3---4---5---6
- $\leftarrow 1 \leftarrow 2$ Start
- Land on 3. So, $5 - 2 = 3$.

Subtracting 2-Digit Numbers (No Borrowing): Arrange under Tens (T) and Units (U).

Subtract Units first, then Tens.

- *Example:* $45 - 12$ T U 4 5
- $12 \text{ ————— } 33$
- **Step 1 (Units):** $5 - 2 = 3$. Write 3 under Units.
- **Step 2 (Tens):** $4 - 1 = 3$. Write 3 under Tens.
- *Answer:* 33. (We have 3 Tens and 3 Units left).

EVALUATION

1. What does the sign "-" mean?
2. Calculate $8 - 3$.
3. Calculate $9 - 0$. (Taking zero away changes nothing).
4. Find the difference between 10 and 4. ($10 - 4 = ?$)
5. Solve: $25 - 11$. (Set it out).
6. Solve: $68 - 24$.

ASSIGNMENT

1. $7 - 5 = ?$
 2. Arrange and solve: $56 - 23$.
 3. Mummy has 8 eggs. She used 3 to make an omelette. How many eggs remain? Write the number sentence.
-

WEEK 10: SUBTRACTION (PART 2).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Subtract 2-digit numbers involving borrowing (renaming).
- Check subtraction answers using addition.
- Solve real-life subtraction word problems involving money or items.

CONTENT

32 - 18

Tens	Units
3 2	12
- 1	- 8
2 - 1 = 1	12 - 8 = 4
1	4
14	

Borrow 1 ten = 10 units

Subtraction with Borrowing: Sometimes the top Unit is smaller than the bottom Unit. We cannot take a big number from a small number directly. We must **borrow** 1 Ten from the Tens column and turn it into 10 Units.

- *Example:* $32 - 18$ T U 2 12 <-- Borrowed: 3 becomes 2, 2 becomes 12. $\begin{array}{r} 32 \\ - 18 \\ \hline \end{array}$
- $18 \text{ ————— } 14$
- **Step 1 (Units):** Look at $2 - 8$. We cannot do $2 - 8$.
- **Borrow 1 Ten** from the 3 in the Tens column. Cross out the 3 and write 2 above it.

- The borrowed 1 Ten is equal to 10 Units. Add these 10 Units to the existing 2 Units.
- So, 2 becomes **12**.
- **Step 2 (Units):** Now do $12 - 8 = 4$. Write 4 under Units.
- **Step 3 (Tens):** The 3 is now 2 (after borrowing).
- Do $2 - 1 = 1$. Write 1 under Tens.
- *Answer:* 14.

Checking Subtraction with Addition: Addition is the opposite of subtraction. To check, add the answer (difference) to the number you subtracted. You should get the number you started with.

- If $10 - 4 = 6$, then check: $6 + 4$ must equal 10.
- Check our example: $32 - 18 = 14$. Is it true? $14 + 18 = 32$. Yes!

3. Real-Life Problems:

- **Money:** I have ₦50. I spent ₦20 on a toy car. How much is left?
- $50 - 20 = ₦30$. I have ₦30 left.
- **Distance:** I need to walk 20 steps to reach the tree. I have walked 10 steps. How many more steps do I need?
- $20 - 10 = 10$ steps more.

EVALUATION

1. Solve: $41 - 19$. (Remember to borrow).
2. Solve: $50 - 25$.
3. Solve: $33 - 14$.
4. Check this answer: Is $10 - 3 = 7$ correct? Show by adding $7 + 3$. What do you get?
5. Tunde has ₦90. He buys a sweet for ₦10. How much change does he have? ($₦90 - ₦10 = ?$)

6. There are 22 birds on a tree. 5 flew away. How many are left?

ASSIGNMENT

1. Set out and solve: $42 - 25$. (Watch the Units: $2 - 5$?).
 2. Set out and solve: $61 - 48$.
 3. A bus has 30 seats. 15 people are sitting. How many seats are empty?
-



WEEK 11: REVISION.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Recall numbers 1--200 and their place values.
- Identify and draw fractions ($\frac{1}{2}$, $\frac{1}{4}$, $\frac{3}{4}$).
- Solve addition problems with carrying.
- Solve subtraction problems with borrowing.

CONTENT

Number Revision: We practice counting 1--200 aloud as a class. We review place value houses: Hundreds, Tens, Units.

- *Example:* In 142, the 1 is Hundreds (100), 4 is Tens (40), 2 is Units (2).
- We review comparison symbols $<$, $>$, $=$. Quick game: Teacher says "67 $\square$ 76", pupils say "less than" and show sign.

2. Fraction Revision: We draw shapes on the board and shade parts.

- *Example:* Draw a rectangle. Divide it into 4 equal parts. Shade 1 part = $\frac{1}{4}$. Shade 3 parts = $\frac{3}{4}$.
- *Key question:* Which is bigger, $\frac{1}{2}$ of a cake or $\frac{1}{4}$ of the same cake?

3. Operations Revision: We practice one harder addition (with carrying) and one harder subtraction (with borrowing) on the board together.

- *Example 1 (Addition):* $28 + 15$ 1 $\leftarrow$ Carry 2 8
- $15 \text{ ————— } 43$
- *Example 2 (Subtraction):* $40 - 12$ T U 3 10 $\leftarrow$ Borrow 4 0
- $12 \text{ ————— } 28$

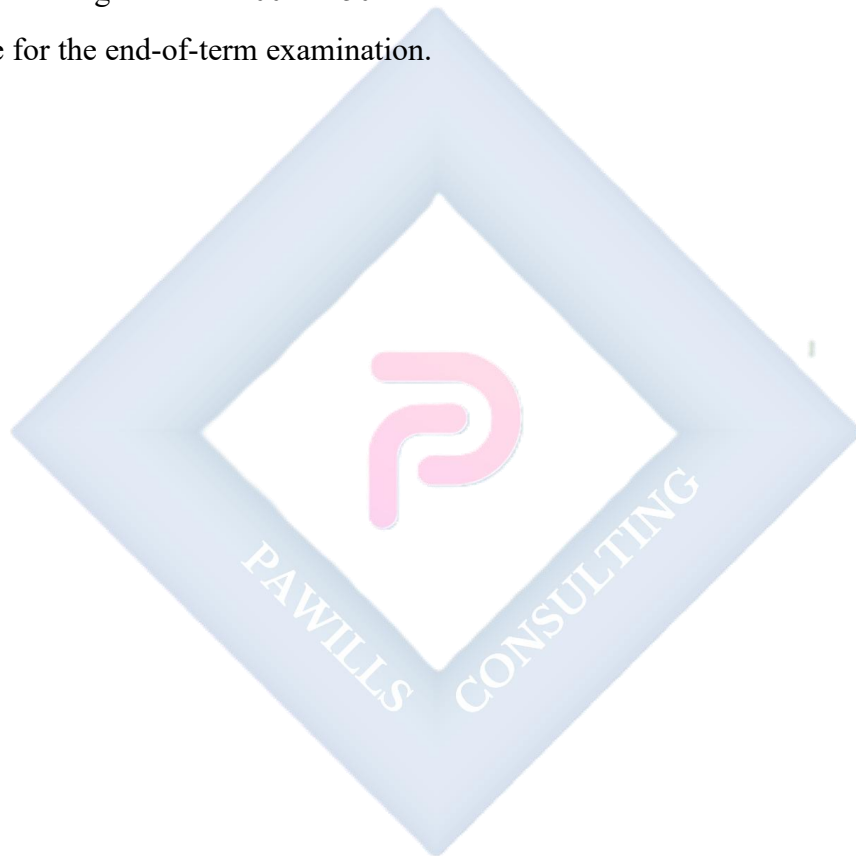
EVALUATION

1. Write 199 in words.

2. Which is bigger: 105 or 150? (Use the correct symbol).
3. Add $17 + 14$. (Show if you need to carry).
4. Subtract $31 - 19$. (Show if you need to borrow).
5. Draw a circle and show $\frac{1}{4}$ by shading one part.

ASSIGNMENT

1. Study your notes from Week 1 to Week 10.
2. Practice writing numbers 100 to 150 in words.
3. Prepare for the end-of-term examination.



PRIMARY 2 MATHEMATICS SCHEME OF WORK

SECOND TERM

Week	Topic	Breakdown (Subtopics)
1	Multiplication (Part 1)	• Meaning of multiplication as repeated addition • Introduction to multiplication tables of 2, 3, 4, and 5 • Multiplying using groups and arrays • Solving simple multiplication problems • Real-life examples (2 bags of 5 oranges each)
2	Multiplication (Part 2)	• Multiplication tables of 6, 7, 8, 9, 10 • Commutative property of multiplication • Multiplying 2-digit by 1-digit numbers • Word problems involving multiplication • Practical multiplication games
3	Open Sentences (Algebraic Expressions)	• Meaning of open sentences • Finding missing numbers: $\square + 4 = 7$ • Using inverse operations to find missing values • Completing number sentences with +, -, $\times$, $\div$ • Word problems with missing numbers
4	Money (Part 1)	• Identifying Nigerian currency: coins and notes • Denominations of Naira and Kobo • Counting money (coins and notes) • Adding and subtracting money • Making change in buying and selling situations

5	MID-TERM EXAMINATION	• Assessment of Weeks 1–4
6	MID-TERM BREAK	• Holiday / Rest
7	Money (Part 2)	• Word problems involving money • Shopping activities and budgeting • Real-life applications of money handling • Understanding saving and spending
8	Length	• Meaning of length • Measuring length with non-standard and standard units (cm, m) • Tools for measuring length (ruler, tape) • Comparing lengths: longer, shorter, equal • Estimating and recording length
9	Time (Part 1)	• Reading time to the hour and half hour • Clock features: face, hands, numbers • Morning, afternoon, evening • Days of the week and months of the year • Activities related to time (daily schedule)
10	Time (Part 2)	• Reading and writing time to quarter past and quarter to • Identifying a.m. and p.m. • Simple duration problems (how long an activity takes) • Word problems on time
11	REVISION	• Review of: Multiplication, open sentences, money, length, time
12	EXAMINATION	• End of Term Assessment
13	CLOSING/VACATION	• Term ends

PRIMARY 2 MATHEMATICS LESSON NOTES

SECOND TERM

WEEK 1: MULTIPLICATION (PART 1).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define multiplication as repeated addition.
- Recite the multiplication tables of 2, 3, 4, and 5.
- Calculate simple multiplication problems using grouping or arrays.
- Solve simple real-life problems involving multiplication.

CONTENT



Meaning of Multiplication (Repeated Addition): Multiplication is a fast way of adding the same number many times. The sign for multiplication is $\times$.

- It is read as "**times**" or "**multiplied by.**"

- If you have 3 apples in one basket and 3 apples in another basket, you can add them: $3 + 3 = 6$.
- Or you can multiply: **2 groups of 3** is $2 \times 3 = 6$.
- Think: "**2 times 3**" means "**3, two times.**"

Multiplication Tables (2, 3, 4, and 5): We learn tables to help us calculate quickly. Let's chant together.

- **2 Times Table:** $2 \times 1 = 2$ (One pair is 2); $2 \times 2 = 4$ (Two pairs are 4); $2 \times 3 = 6$ (Three pairs are 6, like insect legs!); $2 \times 4 = 8$; $2 \times 5 = 10$; $2 \times 6 = 12$
- **3 Times Table:** $3 \times 1 = 3$; $3 \times 2 = 6$ ($3 + 3$); $3 \times 3 = 9$; $3 \times 4 = 12$; $3 \times 5 = 15$
- **4 Times Table:** (Think of legs on animals!) $4 \times 1 = 4$ (One cat has 4 legs); $4 \times 2 = 8$ (Two cats have 8 legs); $4 \times 3 = 12$; $4 \times 4 = 16$; $4 \times 5 = 20$
- **5 Times Table:** (Think of fingers on one hand!) $5 \times 1 = 5$; $5 \times 2 = 10$ (Two hands have 10 fingers); $5 \times 3 = 15$; $5 \times 4 = 20$; $5 \times 5 = 25$

Multiplying Using Groups and Arrays: We can use drawings to find the answer. An **array** is a neat arrangement in rows and columns.

- **Example: 3×4**
- Draw 3 circles (groups). Put 4 dots in each circle.
- Group 1:
- Group 2:
- Group 3:
- Count all the dots: $4 + 4 + 4 = 12$.
- So, $3 \times 4 = 12$.
- **Or as an array:** 3 rows with 4 dots in each row.
-

- Count: 12 dots.

4. Real-Life Examples:

- **Problem:** Mummy bought 2 bags. Each bag has 5 oranges. How many oranges are there?
- *Working:* 2 bags $\times$ 5 oranges = 10 oranges.
- *Number sentence:* $2 \times 5 = 10$.
- **Problem:** A car has 4 wheels. How many wheels do 2 cars have?
- *Working:* 2 cars $\times$ 4 wheels = 8 wheels.
- *Number sentence:* $2 \times 4 = 8$.

EVALUATION

1. What is the sign for multiplication?
2. Fill in the blank: $2 + 2 + 2$ is the same as $3 \times$ _____.
3. Calculate 2×5 .
4. Calculate 4×3 .
5. If one spider has 8 legs, how many legs do 0 spiders have? (Zero property: anything times 0 is 0).
6. Draw 3 groups of 2 balls () and write the total number as a multiplication.
7. Complete: $5 \times 2 =$ _____.

ASSIGNMENT

1. Write out the 2 times table from 2×1 to 2×12 .
 2. Draw groups to show 3×3 . (How many groups? How many in each group?)
 3. A tricycle has 3 wheels. How many wheels do 3 tricycles have? Write the number sentence.
-

WEEK 2: MULTIPLICATION (PART 2).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Recite the multiplication tables of 6 to 10.
- Demonstrate that changing the order of numbers does not change the answer (Commutative Property).
- Multiply a 2-digit number by a 1-digit number.
- Solve word problems involving multiplication.

CONTENT



Multiplication Tables (6--10): We continue learning our tables. Let's use patterns to help.

- **6 Times Table:** (Think of eggs in a half-dozen carton!) $6 \times 1 = 6$; $6 \times 2 = 12$; $6 \times 3 = 18$; $6 \times 4 = 24$; $6 \times 5 = 30$
- **7 Times Table:** (Days in a week!) $7 \times 1 = 7$; $7 \times 2 = 14$ (2 weeks); $7 \times 3 = 21$ (3 weeks); $7 \times 4 = 28$; $7 \times 5 = 35$
- **8 Times Table:** (Legs on spiders?) $8 \times 1 = 8$; $8 \times 2 = 16$; $8 \times 3 = 24$; $8 \times 4 = 32$; $8 \times 5 = 40$
- **9 Times Table:** (Fingers? $9 \times 1 = 9$, $9 \times 2 = 18$) $9 \times 1 = 9$; $9 \times 2 = 18$; $9 \times 3 = 27$; $9 \times 4 = 36$; $9 \times 5 = 45$

- **10 Times Table:** (The easiest! Just add a zero.) $10 \times 1 = 10$; $10 \times 2 = 20$; $10 \times 3 = 30$; $10 \times 4 = 40$; $10 \times 5 = 50$

Commutative Property (Order Doesn't Matter): In multiplication, the order of numbers does not change the answer. We can swap them.

- $2 \times 3 = 6$ and $3 \times 2 = 6$.
- $4 \times 5 = 20$ and $5 \times 4 = 20$.
- Array for 2×3 : (Both have 6 dots)
- Array for 3×2 :
- This means if you forget 8×2 , you can think of 2×8 , which is easier.

3. Multiplying 2-Digit by 1-Digit Numbers: We use our place value columns (Tens and Units) and multiply each part.

- *Example:* 12×2 T U 12×2 ——— 24
- **Step 1 (Units):** $2 \times 2 = 4$. Write 4 under Units.
- **Step 2 (Tens):** $2 \times 1 = 2$. Write 2 under Tens.
- *Answer:* 24.
- *Example:* 21×3 T U 21×3 ——— 63
- **Units:** $3 \times 1 = 3$.
- **Tens:** $3 \times 2 = 6$.
- *Answer:* 63.

4. Word Problems:

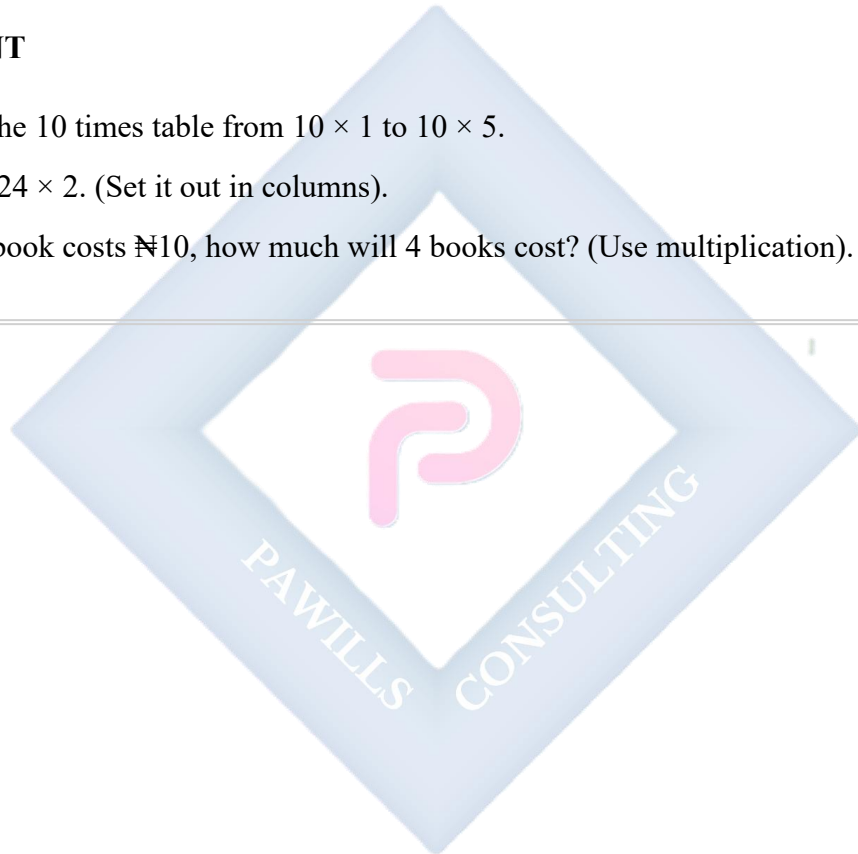
- **Problem:** A classroom has 10 benches. 2 pupils sit on each bench. How many pupils are there?
- *Working:* 10 benches $\times$ 2 pupils = 20 pupils.
- *Number sentence:* $10 \times 2 = 20$.

EVALUATION

1. What is 6×3 ?
2. Is 5×3 the same as 3×5 ? (Yes/No)
3. Solve: 10×5 .
4. Multiply 13 by 2. ($13 \times 2 = ?$)
5. Multiply 21 by 3.
6. A week has 7 days. How many days are in 2 weeks? Write a multiplication sentence.

ASSIGNMENT

1. Write the 10 times table from 10×1 to 10×5 .
2. Solve: 24×2 . (Set it out in columns).
3. If one book costs ₦10, how much will 4 books cost? (Use multiplication).



WEEK 3: OPEN SENTENCES (ALGEBRAIC EXPRESSIONS).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Explain the meaning of an open sentence.
- Find the missing number in addition and subtraction sentences.
- Find the missing number in multiplication sentences.
- Solve simple word problems with missing numbers.

CONTENT



Meaning of Open Sentences: An open sentence is like a puzzle with a missing piece. It has a missing number shown by a box $\square$ or a letter. We need to find the number that makes the sentence true (equal on both sides).

2. Finding Missing Numbers (Addition and Subtraction): We can think: "What number makes this true?" or use the **inverse operation** (the opposite).

- **Example:** $\square + 3 = 7$
- *Ask:* "What do I add to 3 to get 7?"
- Count up from 3 to 7: 4, 5, 6, 7. That's 4 more.

- So, $\square = 4$.
- *Inverse Method:* The opposite of $+$ is $-$. So, $7 - 3 = 4$.
- **Example:** $10 - \square = 6$
- *Ask:* "10 take away what gives 6?"
- Think: $10 - 6 = 4$.
- So, $\square = 4$.
- *Inverse Method:* $10 - 6 = 4$, or $6 + 4 = 10$.

3. Finding Missing Numbers (Multiplication):

- **Example:** $\square \times 2 = 10$
- *Ask:* "How many times do I count 2 to get 10?"
- Count in 2s: 2, 4, 6, 8, 10. That's 5 times.
- So, $\square = 5$.
- *Inverse Method:* The opposite of $\times$ is $\div$. So, $10 \div 2 = 5$.

4. Word Problems:

- **Problem:** I have some mangoes. I ate 2, and now I have 4 left. How many did I start with?
- *Sentence:* $\square - 2 = 4$.
- *Working:* The opposite of $- 2$ is $+ 2$. So, $4 + 2 = 6$.
- I started with 6 mangoes.

EVALUATION

1. Find the missing number: $5 + \square = 9$.
2. Find the missing number: $\square - 3 = 5$.
3. Complete: $2 \times \square = 8$.
4. $10 + 5 = \square$. (This one is easy!)

5. $\square + 10 = 20$.

6. If $\square \times 3 = 9$, what is inside the box?

ASSIGNMENT

1. Find the missing numbers: a) $6 + \square = 10$ b) $15 - \square = 10$
2. Think of a number. If you add 2 to it, the answer is 7. What is the number? Write the open sentence and solve it.

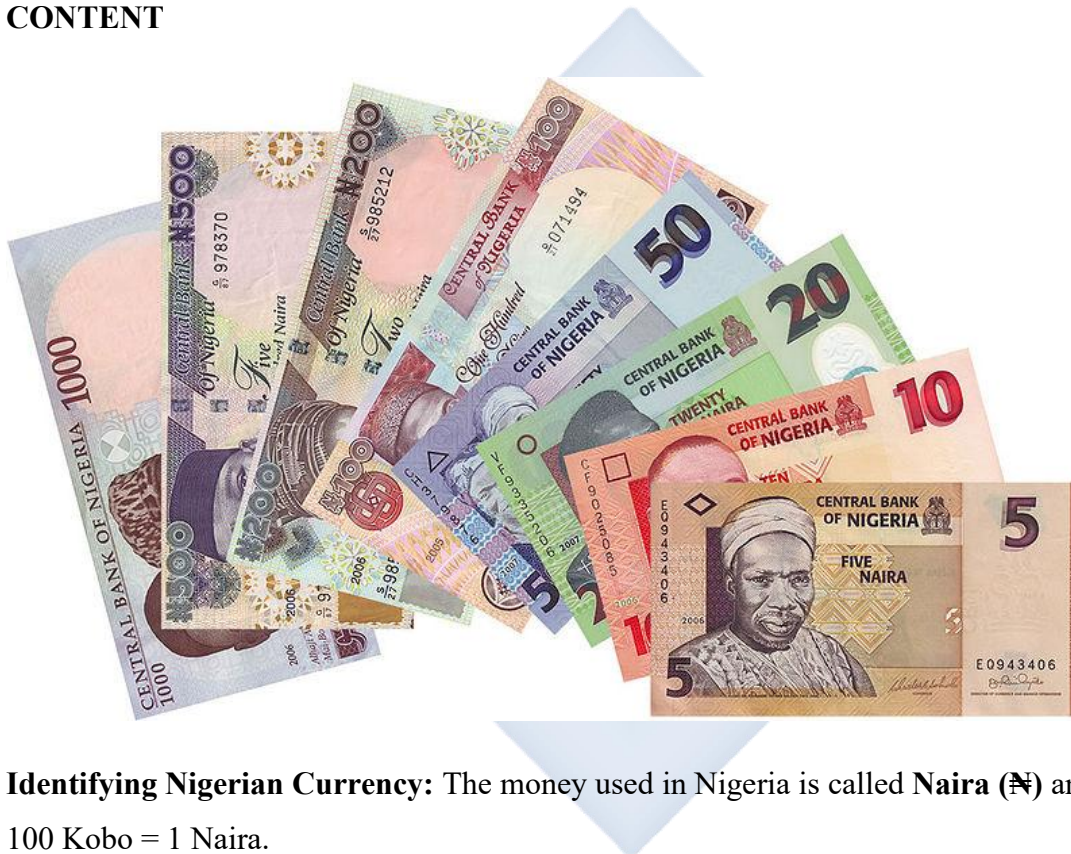


WEEK 4: MONEY (PART 1).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Identify Nigerian coins and bank notes.
- List the different denominations of Naira and Kobo.
- Count combinations of notes to find total value.
- Add simple sums of money.

CONTENT



Identifying Nigerian Currency: The money used in Nigeria is called **Naira (₦)** and **Kobo (k)**.

100 Kobo = 1 Naira.

- **Coins:** 50k, ₦1, ₦2. Coins are usually hard and round.
- **Notes (Banknotes):** ₦5, ₦10, ₦20, ₦50, ₦100, ₦200, ₦500, ₦1000. Notes are paper.

We can recognize notes by their color and picture:

- ₦5: Light blue
- ₦10: Orange/Brown
- ₦20: Green

- ₦50: Blue/Purple
- ₦100: Brown/Blue
- ₦200: Green/Orange
- ₦500: Purple/Red
- ₦1000: Light Blue/Grey

Counting Money: We count money by adding the values of the notes and coins together.

- One ₦10 note = ₦10.
- Two ₦20 notes = $20 + 20 = ₦40$.
- One ₦50 note and one ₦10 note = $50 + 10 = ₦60$.
- *Example:* If I have ₦5, ₦10, and ₦20, my total is $5 + 10 + 20 = ₦35$.

Adding Money: We add money just like we add normal numbers. We can write it in columns (₦ first).

- *Example:* Add ₦25 and ₦10. T U 2 5 (₦25)
- 1 0 (₦10) ————— 3 5 (₦35)
- *Answer:* ₦35.

Making Change (Introduction): Change is the money you get back when you pay more than the price of an item.

- If a sweet costs ₦5 and you give the seller ₦10, the seller must give you money back.
- Amount to return: $10 - 5 = ₦5$.
- So, your change is ₦5.

EVALUATION

1. What is the symbol for Naira?
2. Mention three Nigerian bank notes you know.
3. How much is two ₦10 notes? ($₦10 + ₦10 = ?$)
4. Add ₦20 + ₦20.

5. If I have ₦50 and ₦10, how much money do I have altogether?
6. Which is bigger: ₦100 or ₦200?

ASSIGNMENT

1. Draw the sign for Naira (₦) and for Kobo (k).
 2. Calculate: ₦15 + ₦10.
 3. List the colors of the ₦20 and ₦50 notes.
-



WEEK 5 & 6: MID-TERM EXAMINATION AND MID-TERM BREAK.

(Note: No lesson notes are required for these weeks as they are for assessment and holiday.)



WEEK 7: MONEY (PART 2).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Solve simple word problems involving money.
- Calculate the total cost of items in a shopping list.
- Determine change to be collected after buying items.
- Explain the importance of saving money.

CONTENT



Shopping: Calculating Total Cost: When we go to the shop, we add the prices of everything we buy to find the total cost.

- *Example:* A pencil costs ₦10. An eraser costs ₦5. What is the total?
- *Working:* ₦10 + ₦5 = ₦15.
- The total cost is ₦15.

Word Problems on Change: Change is the money you get back when you pay more than the price. To find change, subtract the price from the amount you paid.

- *Problem:* You have ₦50. You buy a biscuit for ₦30. How much change will you get?
- *Working:* $₦50 - ₦30 = ₦20$.
- Your change is ₦20.

3. Saving and Spending:

- **Spending:** Using money to buy things we **need** (like food, books, uniform) or things we **want** (like toys, sweets).
- **Saving:** Keeping money for later instead of spending it all. We can save in a piggy bank or at a real bank.
- **Why Save?:** Saving helps us buy big things later (like a bicycle), or have money for an emergency (like when we are sick).

EVALUATION

1. A book costs ₦40. I have only ₦20. How much more money do I need to buy the book? (Hint: Subtract).
2. Calculate the total: $₦20 + ₦20 + ₦10$.
3. If you buy sweets for ₦10 and pay with ₦20, what is your change?
4. Why should we save money?
5. Musa has ₦100. He lost ₦20. How much does he have left?

ASSIGNMENT

1. Ask your parents the price of a loaf of bread and write it down.
 2. You have ₦50. You want to save ₦20. How much can you spend now? ($₦50 - ₦20 = ?$)
 3. Calculate: $₦100 - ₦40$.
-

WEEK 8: LENGTH.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define length as the measurement of how long or tall something is.
- Identify tools for measuring length (ruler, tape measure).
- Measure objects using non-standard units (hand span, foot).
- Measure objects using standard units (centimeters and meters).
- Compare lengths using "longer," "shorter," or "taller."

CONTENT



Meaning of Length: Length tells us how long an object is from one end to the other. It also tells us how tall someone or something is. We measure length in straight lines.

2. Tools for Measuring:

- **Ruler:** Used for short things like a pencil, book, or your finger. It measures in **centimeters (cm)**.
- A ruler: |---|---|---|---| (each cm marked)

- **Tape Measure:** A soft, flexible ruler. Used for long or curved things like a table, waist, or height. It measures in **meters (m)** and **centimeters (cm)**.
- 1 meter (m) = 100 centimeters (cm).

3. Non-Standard vs. Standard Units:

- **Non-Standard:** Using body parts. This is not the same for everyone.
- **Hand Span:** The width of your open hand from thumb to little finger.
- **Foot:** The length of your foot.
- *Problem:* My hand is smaller than your hand, so if we both measure the same table, we will get different numbers! That's confusing.
- **Standard:** Using a ruler or tape measure (cm and m).
- This is better because 1 cm is always the same size everywhere in the world.

4. Comparing Lengths:

- A pencil is **shorter** than a ruler.
- A teacher is **taller** than a pupil.
- A snake is **longer** than a worm.
- We use the words: longer/shorter (for objects), taller/shorter (for people/height).

EVALUATION

1. What tool do we use to measure a straight line in our book?
2. Which is better to use: a hand span or a ruler? Why?
3. What unit do we see on a ruler? (cm or kg?)
4. Measure the length of your exercise book using your ruler. Write the number in cm.
5. Who is taller: You or your teacher?
6. Is a car longer or shorter than a bicycle?

ASSIGNMENT

1. Measure the length of your pencil in cm.
 2. Measure the width of your table using your hand span. How many hand spans?
 3. Draw a line that is 5cm long. (Use your ruler).
-

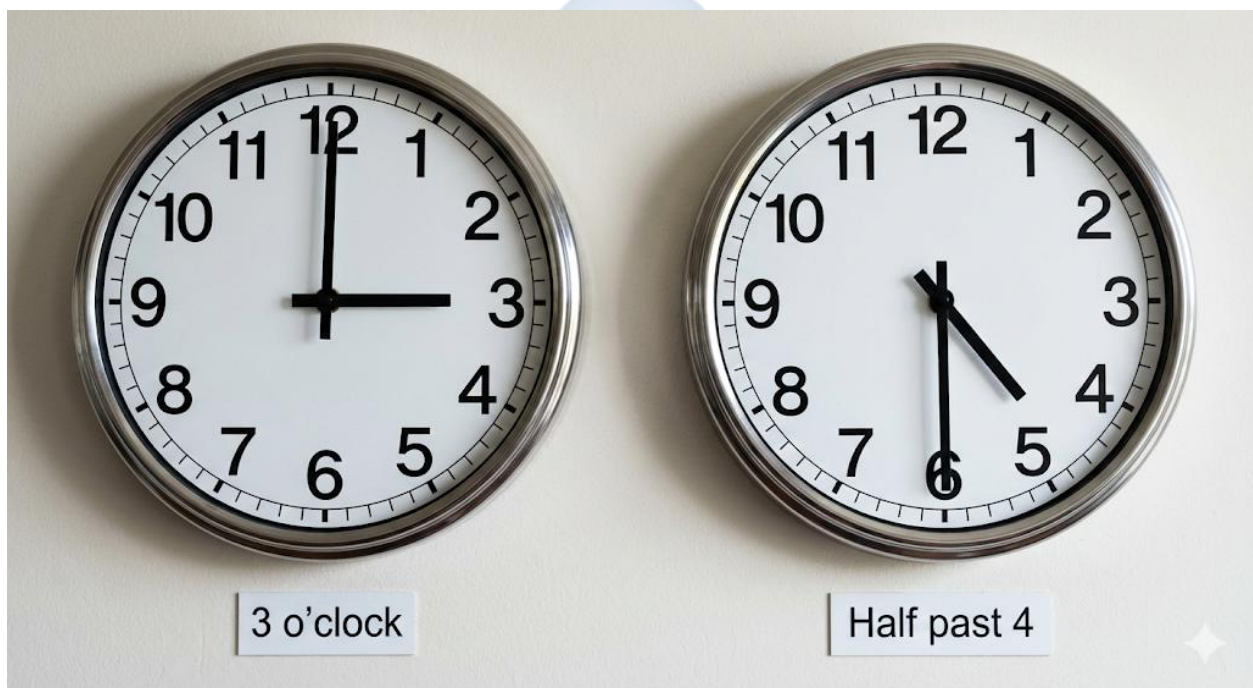


WEEK 9: TIME (PART 1).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Identify the parts of a clock (face, long hand, short hand).
- Read time to the exact hour (e.g., 2 o'clock).
- Read time to the half hour (e.g., half past 2).
- List the days of the week and months of the year.

CONTENT



Features of a Clock: A clock tells us the time of day.

- **Face:** The front of the clock with numbers 1 to 12 in a circle.
- **Long Hand:** The **minute hand**. It is long and thin. It moves fast and points to the minutes.
- **Short Hand:** The **hour hand**. It is short and fat. It moves slowly and points to the hour.

Reading Time (O'clock): When the **Long Hand (minute hand)** is pointing straight up at **12**, it is an exact hour. We look at where the **Short Hand (hour hand)** is pointing.

- Short hand at 3, Long hand at 12 = **3 o'clock**.
- Short hand at 8, Long hand at 12 = **8 o'clock**.

Reading Time (Half Past): When the **Long Hand (minute hand)** is pointing straight down at **6**, it is **half past** the hour. The Short Hand will be halfway between two numbers.

- Short hand between 2 and 3, Long hand at 6 = **Half past 2**.
- Think: 30 minutes have passed since the hour.

4. Days and Months:

- There are **7 days** in a week: Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday.
- There are **12 months** in a year: January, February, March, April, May, June, July, August, September, October, November, December.

EVALUATION

1. Where does the long hand point for "o'clock"?
2. Where does the long hand point for "half past"?
3. If the short hand is on 4 and the long hand is on 12, what is the time?
4. How many days are in a week?
5. Which month comes after March?
6. Show 9 o'clock on your clock drawing.

ASSIGNMENT

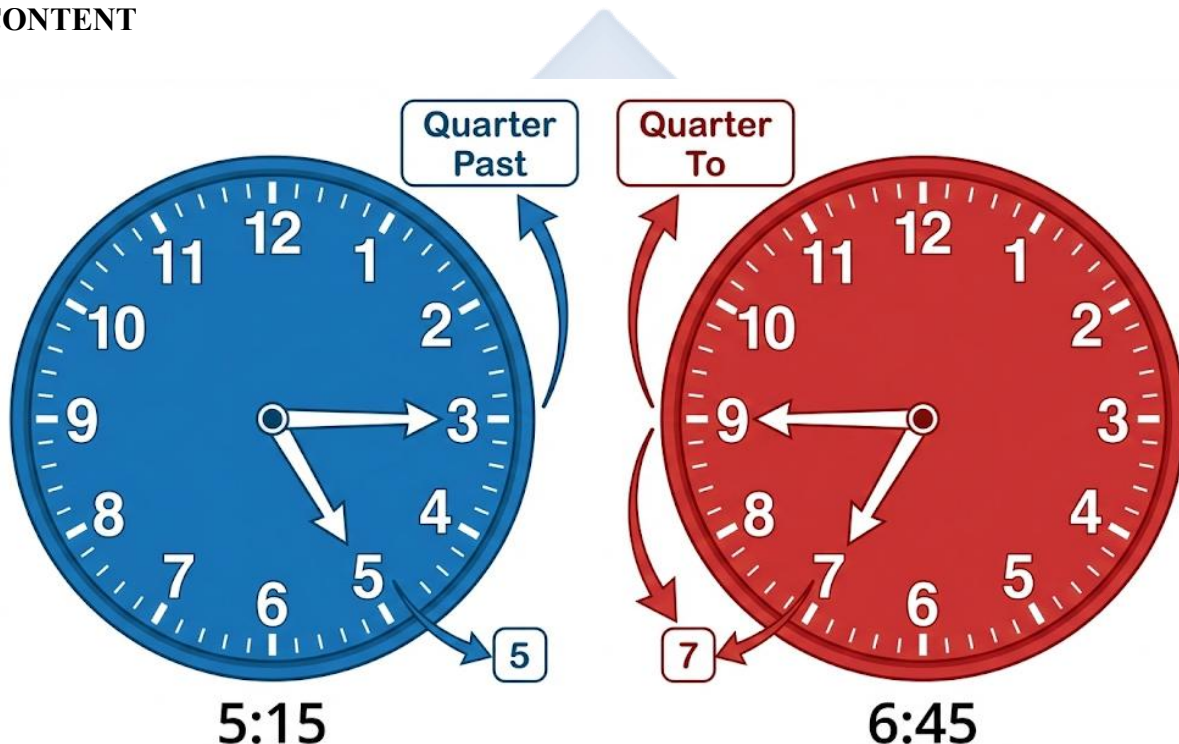
1. Draw a clock showing **6 o'clock**.
2. Write the names of all the days of the week in order.
3. What time do you come to school? (Ask your teacher if you don't know).

WEEK 10: TIME (PART 2).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Read time to "Quarter Past" and "Quarter To."
- Differentiate between a.m. (morning) and p.m. (afternoon/evening).
- Solve simple duration problems (how long something takes).
- Solve simple word problems involving time.

CONTENT



Quarter Past and Quarter To:

- **Quarter Past (15 minutes past):** The Long Hand points to 3.
- *Example:* Quarter past 1 (Long hand at 3, Short hand just past 1).
- "Quarter" means $\frac{1}{4}$ of an hour (60 minutes). $\frac{1}{4}$ of 60 minutes = 15 minutes.
- **Quarter To (15 minutes to the next hour):** The Long Hand points to 9.
- *Example:* Quarter to 1 (Long hand at 9, Short hand almost at 1).

- It means 15 minutes before 1 o'clock.

2. A.M. and P.M.:

- **a.m. (ante meridiem):** Used for times from **midnight (12 a.m.) to midday (11:59 a.m.)**.
- This is the **morning**.
- e.g., 8:00 a.m. (Assembly time), 6:00 a.m. (Wake up time).
- **p.m. (post meridiem):** Used for times from **midday (12 p.m.) to midnight (11:59 p.m.)**.
- This is the **afternoon and evening**.
- e.g., 2:00 p.m. (Closing time), 8:00 p.m. (Sleeping time).

Simple Duration: Duration is how long an activity lasts from start to finish. To find it, we can subtract the start time from the end time.

- If you start eating at 2 o'clock and finish at 3 o'clock, you spent **1 hour** eating.
- Calculation: $3 - 2 = 1$ hour.

4. Word Problems:

- **Problem:** School starts at 8 o'clock. Tolu is 1 hour late. What time does Tolu arrive?
- *Working:* $8 \text{ o'clock} + 1 \text{ hour} = 9 \text{ o'clock}$.
- Tolu arrives at 9 o'clock.

EVALUATION

1. When the long hand is on 3, do we say "Quarter Past" or "Quarter To"?
2. When the long hand is on 9, what do we say?
3. Do we eat breakfast at 7 a.m. or 7 p.m.?
4. Do we sleep at night at 9 a.m. or 9 p.m.?
5. If a cartoon starts at 4 o'clock and ends at 5 o'clock, how long was the cartoon?
6. Draw a clock showing "Quarter past 5."

ASSIGNMENT

1. Write down one thing you do in the **a.m.** and one thing in the **p.m.** (e.g., a.m. - brush teeth, p.m. - do homework).
 2. If you walk for 2 hours starting at 1 o'clock, when will you stop? ($1 + 2 = ?$)
 3. Draw a clock showing **Quarter to 10**.
-



WEEK 11: REVISION.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Recall multiplication tables 2-10.
- Solve mixed problems on money, length, and time.
- Find missing numbers in open sentences correctly.

CONTENT

Multiplication Review: Quick fire quiz on tables.

- Teacher: " 5×5 ?" Pupils: "25!"
- Teacher: " 3×4 ?" Pupils: "12!"
- Let's chant the 10 times table together.

2. Money and Open Sentences:

- Review adding money: $\text{₦}20 + \text{₦}30 = \text{₦}50$.
- Review finding missing numbers: $5 + \square = 12$. Ask: " $12 - 5 = ?$ "
- Answer: 7.

3. Length and Time Review:

- **Measuring:** Let's measure the teacher's table in hand spans, then with a ruler. Which is more accurate?
- **Reading the Clock:** Look at the clock on the wall. Is it an 'o'clock', 'half past', 'quarter past', or 'quarter to'?

EVALUATION

1. What is 3×3 ?
2. Find $\square$: $10 - \square = 8$.
3. How many kobo make one Naira? (Hint: $100\text{k} = \text{₦}1$).
4. Show 12 o'clock on the clock. (Where are both hands?)
5. Which is longer: a meter or a centimeter? (Remember: $1\text{ m} = 100\text{ cm}$).

ASSIGNMENT

1. Study your notes from Week 1 to Week 10.
2. Practice drawing clocks at home showing: 3 o'clock, half past 7.
3. Prepare for the end-of-term examination.



PRIMARY 2 MATHEMATICS SCHEME OF WORK

THIRD TERM

Week	Topic	Breakdown (Subtopics)
1	Weight	• Meaning of weight (how heavy something is) • Units of weight: gram (g), kilogram (kg) • Measuring using a scale • Comparing objects: heavier, lighter, same weight • Real-life weight estimation activities
2	Capacity	• Meaning of capacity (how much a container holds) • Measuring capacity using cups, bottles, and jugs • Units of capacity: millilitre (ml), litre (L) • Comparing capacities: full, half-full, empty • Word problems on capacity
3	Area (Introduction)	• Meaning of area (space covered by a surface) • Measuring area using square units • Comparing areas of objects (bigger/smaller) • Practical activities: measuring table or book covers • Word problems on area
4	2-Dimensional Shapes	• Identifying and naming 2D shapes: circle, square, rectangle, triangle • Properties of 2D shapes (sides, corners) • Drawing and tracing shapes • Sorting and grouping shapes • Shapes in our environment
5	MID-TERM EXAMINATION	• Assessment of Weeks 1–4
6	MID-TERM BREAK	• Holiday / Rest

7	3-Dimensional Shapes	- Identifying and naming 3D shapes: cube, cuboid, cylinder, sphere, cone - Properties of 3D shapes: faces, edges, vertices - Relationship between 2D and 3D shapes - Finding 3D shapes in our environment
8	Data Collection (Everyday Statistics)	- Meaning of data (information) - Collecting and recording simple data - Using tally marks and pictograms - Reading and interpreting pictorial data - Word problems on data
9	Review of Geometry Concepts	- Comparing 2D and 3D shapes - Revision of weight, capacity, and area - Applying geometry concepts in real-life contexts - Hands-on measuring and drawing activities
10	Applied Problem Solving (All Concepts)	- Mixed operations with money, measurement, and time - Real-life scenarios combining different mathematical concepts - Solving practical quantitative reasoning problems
11	REVISION	Comprehensive review of all third-term topics
12	EXAMINATION	End of Term Assessment
13	CLOSING/VACATION	Term ends

PRIMARY 2 MATHEMATICS LESSON NOTES

THIRD TERM

WEEK 1: WEIGHT.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define weight as the measurement of how heavy or light an object is.
- Identify grams (g) and kilograms (kg) as the standard units of weight.
- Compare objects using the terms "heavier," "lighter," or "same weight."
- Estimate the weight of common classroom objects.

CONTENT



Meaning of Weight: Weight tells us how heavy something is when we lift it. It is the force that pulls objects down to the ground.

- Some things are very heavy (like a bag of rice, a big stone, a car).
- Some things are very light (like a feather, a piece of paper, a leaf).
- When we carry something heavy, it is hard to lift. When we carry something light, it is easy to lift.

Units of Weight (Gram and Kilogram): We use special units to measure weight so everyone understands the same amount.

- **Gram (g):** We use grams for light things.
- *Examples:* A pencil (about 10g), an eraser (about 5g), a sweet (about 2g).
- **Kilogram (kg):** We use kilograms for heavy things.
- *Examples:* A bag of rice (5kg), a big yam (2kg), a pupil (maybe 20kg).
- 1000 grams make 1 kilogram.
- **1 kg = 1000 g**

Measuring Using a Scale: We use a tool called a **Scale** or **Weighing Balance** to measure weight. When we put an object on a scale, the needle moves to show the number in kg or g. In the market, sellers use scales to sell meat, fish, or fruits.

4. Comparing Objects:

- **Heavier:** Weighs more. (A stone is heavier than a feather).
- **Lighter:** Weighs less. (A book is lighter than a table).
- **Same Weight:** Weighs the same amount. (Two identical pencils have the same weight).

Balance Scale: [Stone] ↓ | ↑ [Feather] Heavier Lighter

EVALUATION

1. What does weight measure?
2. Mention one object that is heavy.
3. Mention one object that is light.
4. Which unit do we use for heavy things: gram or kilogram?
5. Is a car heavier or lighter than a bicycle?
6. Which tool do we use to measure weight?

ASSIGNMENT

1. Draw a picture of a scale (a simple balance).
 2. Write down two things in your house that are heavy (use kg if you know).
 3. Write down two things in your house that are light (use g if you can guess).
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WEEK 2: CAPACITY.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define capacity as the amount of liquid a container can hold.
- List units of capacity (millilitre and litre).
- Measure capacity using cups and bottles.
- Compare the capacity of different containers (full, half-full, empty).

CONTENT



Meaning of Capacity: Capacity means how much **space** is inside a container to hold liquid like water, juice, petrol, or sand. It tells us how much a container can **contain**.

- A big bucket has a **large capacity** (can hold a lot).
- A small spoon has a **small capacity** (can hold a little).

2. Units of Capacity:

- **Millilitre (ml):** Used for very small amounts of liquid.
- *Example:* Medicine in a spoon (5 ml), a small juice box (200 ml).

- **Litre (L):** Used for large amounts of liquid.
- *Example:* A bottle of water (1 L), a bucket of water (10 L), a big bottle of soft drink (2 L).
- 1000 millilitres make 1 litre.
- **1 L = 1000 ml**

Measuring Capacity: We can use measuring cups, jugs, or bottles with marks on the side. If a jug holds 4 cups of water, we say the capacity of the jug is 4 cups. We can pour water from a bottle into a **measuring jug** to see how many Litres it is.

4. Comparing Capacities:

- **Full:** The liquid reaches the very top of the container.
- **Half-full:** The liquid is exactly in the middle of the container.
- **Empty:** There is no liquid inside at all.
- A bucket holds **more** water than a cup. (Bucket has greater capacity).
- A spoon holds **less** water than a bottle. (Spoon has smaller capacity).

EVALUATION

1. What is capacity?
2. Which holds more water: a bucket or a cup?
3. Mention one unit we use to measure liquid (ml or L).
4. If a bottle is "empty," what does it mean?
5. How many cups of water do you think can fill a bucket: 2 cups or 20 cups?
6. Draw a cup that is half-full. (Shade the bottom half).

ASSIGNMENT

1. Draw a bottle and label it "1 Litre".
2. Get a cup and a spoon at home. How many spoons of water fill the cup? Write the number. (This is a fun experiment!).
3. Which unit is bigger: Millilitre or Litre?

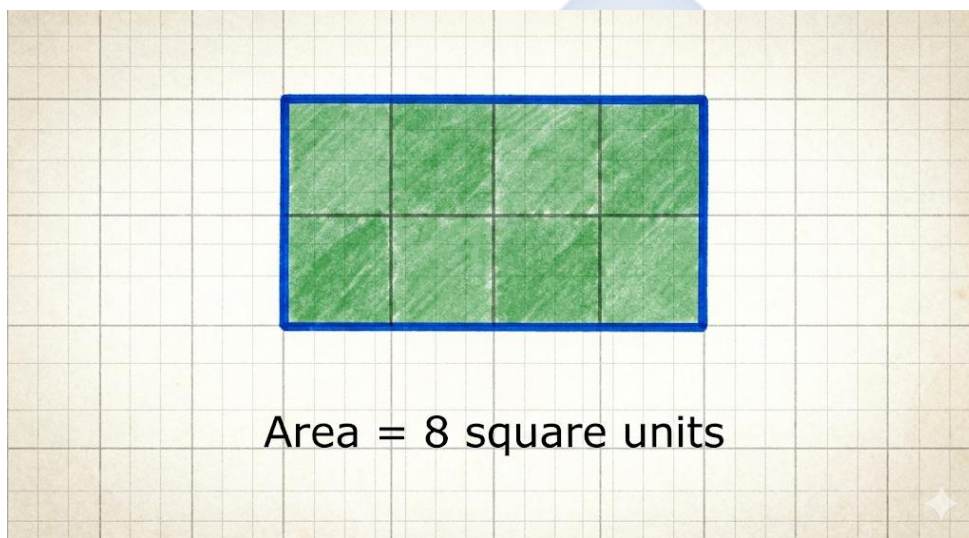


WEEK 3: AREA (INTRODUCTION).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define area as the amount of space covered by a surface.
- Measure area using square units (counting squares).
- Compare the areas of two different flat objects.
- Solve simple problems involving area.

CONTENT



Meaning of Area: Area is the space on top of a flat shape or surface. It tells us **how much surface** an object covers. Area is measured in **square units**.

- The top of your desk has an area.
- The cover of your notebook has an area.
- The floor of the classroom has a large area.

Measuring Area with Square Units: To measure area, we count how many small squares can fit inside the shape **without gaps or overlaps**. If we draw a rectangle on squared paper and it covers 6 small squares, the area is **6 square units**.

- Example shape covering 6 squares:  (Each  is a square unit. Total: 6)

3. Comparing Areas:

- **Bigger Area:** Covers more space. (A teacher's table has a bigger area than a pupil's desk).
- **Smaller Area:** Covers less space. (A stamp has a smaller area than an envelope).

4. Practical Activity: We can use our exercise books (or square tiles) to measure the area of the table.

- *Ask:* "How many books can cover the top of the desk without overlapping?"
- If 10 books cover it, the area is about 10 books.

EVALUATION

1. What is area?
2. Which has a bigger area: a blackboard or an exercise book?
3. If a shape contains 8 small squares, what is its area? (Remember the unit!).
4. Look at your desk and your chair seat. Which surface is bigger?
5. How do we measure area in this lesson? (By counting squares/objects).

ASSIGNMENT

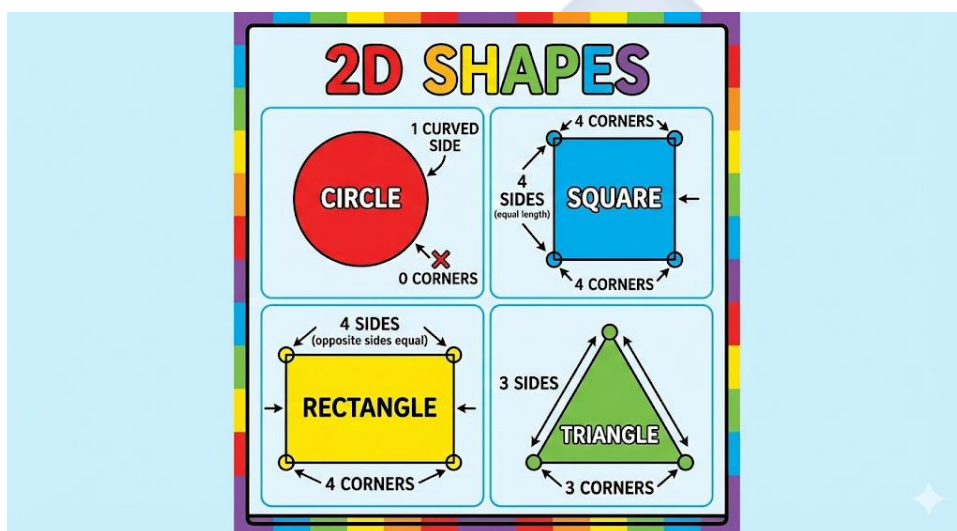
1. Draw a square on a squared page. Draw grid lines inside to make 4 small squares.
 2. Write "Area = 4 square units."
 3. Place your eraser on your notebook. Guess how many erasers can cover the notebook.
Write your guess.
 4. Which has a smaller area: your palm or your foot? (Compare by placing one on top of the other on paper).
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WEEK 4: 2-DIMENSIONAL SHAPES.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Identify and name basic 2D shapes (square, rectangle, triangle, circle).
- Describe the properties of 2D shapes (sides and corners).
- Draw 2D shapes correctly.
- Identify 2D shapes in the environment.

CONTENT



Identifying 2D Shapes: 2D shapes are **flat**. They have length and width but no thickness. We can draw them on paper.

- **Circle:** Round like a ball, a coin, or a clock face. It has one curved side.
- **Triangle:** Has 3 straight sides and 3 pointed corners like a slice of pizza or a road sign.
- **Square:** Box shape. All 4 sides are equal in length. Like a window pane or a chess board square.
- **Rectangle:** Box shape. Two opposite sides are long, and the other two opposite sides are short. Like a door, a book, or a sheet of paper.

2. Properties of Shapes:

- **Side:** The straight line that makes the shape.

- **Corner (Vertex):** The point where two sides meet.

Shape	Sides	Corners
Circle	1	0
Triangle	3	3
Square	4	4
Rectangle	4	4

- **Square vs. Rectangle:** A square is a special rectangle where ALL sides are equal.

Drawing and Tracing: We practice drawing shapes:

- Use a **ruler** for straight lines (triangle, square, rectangle).
- Use a **coin** or **cup** to trace a perfect circle.

4. Shapes in the Environment: Let's look around:

- **Door:** Rectangle.
- **Clock face:** Circle.
- **Floor tile:** Square.
- **Sandwich cut in half:** Triangle.
- **Window:** Square or Rectangle.

EVALUATION

1. Draw a Triangle. How many sides does it have?
2. How many corners does a Circle have?
3. What is the difference between a Square and a Rectangle? (Hint: Look at the sides).
4. Name a shape that has 4 equal sides and 4 corners.
5. Look at the classroom door. What shape is it?
6. How many corners does a Square have?

ASSIGNMENT

1. Draw a simple house using a Triangle (for the roof) and a Square or Rectangle (for the walls).
 2. Write down the number of sides for: Triangle, Square, Circle.
 3. Find one object in your house that is a Circle and write its name.
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WEEK 5 & 6: MID-TERM EXAMINATION AND MID-TERM BREAK.

(Note: No lesson notes are required for these weeks as they are for assessment and holiday.)



WEEK 7: 3-DIMENSIONAL SHAPES.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Identify and name 3D shapes (cube, cuboid, cylinder, sphere, cone).
- Differentiate between flat (2D) and solid (3D) shapes.
- Count the faces and edges of simple 3D shapes.
- Match real-life objects to their 3D shapes.

CONTENT



Identifying 3D Shapes: 3D shapes are **solid**. They are fat, not flat. You can hold them. They have length, width, and **height** (thickness).

- **Cube:** Like a sugar cube, dice, or some gift boxes. All faces are squares.
- **Cuboid:** Like a shoebox, book, or matchbox. Faces are rectangles (or squares).
- **Cylinder:** Like a tin of milk, a pipe, or a drum. It has two flat circle faces and one curved face.
- **Sphere:** Like a football, orange, or marble. Completely round, no edges.
- **Cone:** Like an ice-cream cone, a party hat, or a traffic cone. Has a circular base and a pointed top.

2. Properties (Faces and Edges):

- **Face:** The flat part of the shape.
- **Edge:** The line where two faces meet.
- **Corner (Vertex):** The point where edges meet.

Shape	Faces	Edges	Corners
Cube	6	12	8
Cuboid	6	12	8
Cylinder	3	2	0
Sphere	1	0	0
Cone	2	1	1

3. 2D vs 3D:

- A **Square** is flat (2D). A **Cube** is solid (3D) and is made of 6 squares.
- A **Circle** is flat (2D). A **Sphere** is solid (3D) and is like a 3D circle.

4. Real-life Objects:

- **Tin of tomatoes:** Cylinder.
- **Ball:** Sphere.
- **Dice:** Cube.
- **Matchbox:** Cuboid.
- **Birthday hat:** Cone.

EVALUATION

1. Name this shape: It looks like a ball.
2. Name this shape: It looks like a tin of milk.
3. Which 3D shape looks like a box? (Cube or Cuboid?).
4. How many flat faces does a cylinder have? (Look at the top and bottom).
5. Can a cube roll like a ball? Why or why not? (Think about its faces).
6. Draw a Cone. (A triangle on a circle).

ASSIGNMENT

1. Write the names of three objects in your kitchen and their 3D shapes (e.g., Milk tin - Cylinder).
 2. Draw a Cube. (Hint: Draw a square, then draw another square behind it and connect the corners).
 3. Which shape is an orange? (Circle or Sphere?)
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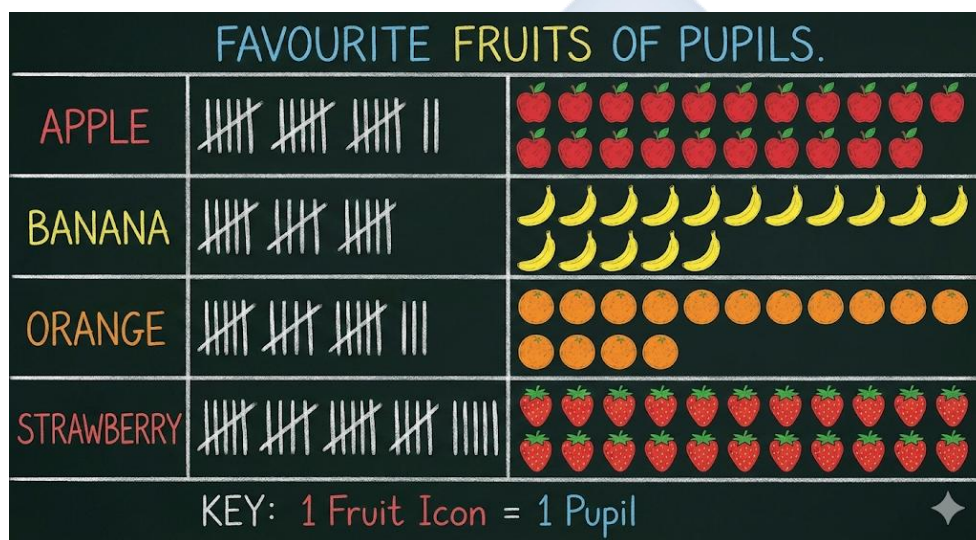


WEEK 8: DATA COLLECTION (EVERYDAY STATISTICS).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define data as information we collect.
- Collect simple data using tally marks.
- Represent data using pictograms (drawings).
- Interpret simple data to answer questions.

CONTENT



Meaning of Data: Data is **information** we collect by counting or asking questions. It tells us "how many" or "how much."

- *Example:* Counting how many boys and girls are in the class.
- *Example:* Finding out the favorite fruit of pupils.

Using Tally Marks: A tally mark is a straight line | used to count one item. We group them in fives for easy counting.

- 1 = |
- 2 = ||
- 3 = |||
- 4 = ||||

- $5 = ||||$ (Four lines with a diagonal line through them). This is called a "**gate**".
- **Tally for 7:** $|||| ||$ ($5 + 2 = 7$)

Pictograms: A pictogram uses **pictures** or **symbols** to show numbers. Each picture stands for one item or a group of items. If we are counting apples, we draw small apple pictures.

- **Ali:** 🍏 🍏 🍏
- **Bola:** 🍏 🍏
- If Ali has 3 apples, we draw 3 apples. If Bola has 2, we draw 2.

4. Reading Data: We look at the data to answer questions.

- Look at the pictogram above. **Who has more apples?** (Ali).
- **How many apples do they have altogether?** $3 + 2 = 5$ apples.

EVALUATION

1. What is data?
2. Draw tally marks for the number 4.
3. Draw tally marks for the number 6. (Remember the gate for 5).
4. If one smiley face (😊) means 1 pupil, how many pupils are shown by 3 smiley faces?
(😊😊😊)
5. Count the number of windows in the class and write it down as a number.
6. Who has more: The person with 5 tally marks or the person with 3 tally marks?

ASSIGNMENT

1. Count the number of chairs in your living room and write the number using tally marks.
2. Draw a pictogram showing:
 - Red Balls: 2 (Draw 2 red circles ●)
 - Blue Balls: 4 (Draw 4 blue circles ●)
3. How many balls are there in total from question 2? (Add them up).

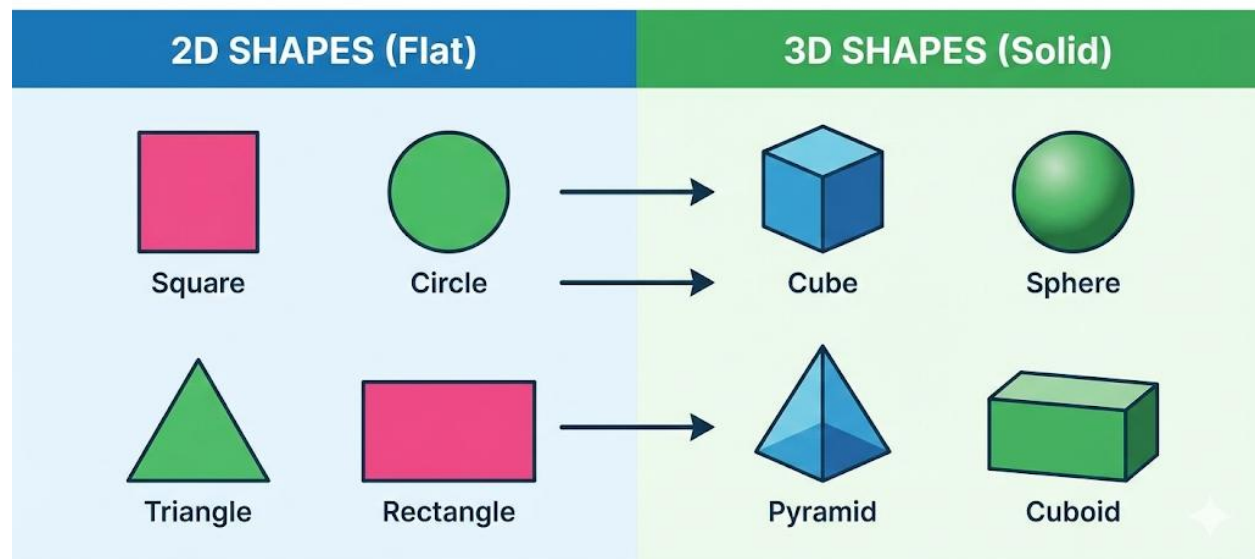
WEEK 9: REVIEW OF GEOMETRY CONCEPTS.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Distinguish clearly between 2D and 3D shapes.
- Recall the units for weight, capacity, and area.
- Measure objects using the correct tools in a practical session.
- Solve mixed word problems on geometry.

CONTENT

2D vs. 3D GEOMETRY REVIEW



Comparing 2D and 3D Shapes: We mix all the shapes together and sort them.

- **2D (Flat):** Triangle, Square, Circle, Rectangle. You cannot hold them, only draw them.
- **3D (Solid):** Cone, Cube, Sphere, Cylinder, Cuboid. You can hold them; they take up space.
- **Activity:** The teacher holds up a shape (or picture), and pupils shout "2D" or "3D".

2. Review of Measurement: Let's recall the units and tools:

- **Weight:** How heavy? **Units:** Kilogram (kg), Gram (g). **Tool:** Scale.

- **Capacity:** How much liquid? **Units:** Litre (L), Millilitre (ml). **Tool:** Measuring Jug/Cup.
- **Area:** How much surface? **Units:** Square units. **Tool:** Counting squares.

3. Practical Activities: We will work in groups:

- **Group A:** Measure the capacity of a small bucket using a cup. How many cups fill it?
- **Group B:** Sort a box of objects into piles of cubes, spheres, and cylinders.
- **Group C:** Compare the weight of two objects (like a stone and a book) by holding one in each hand. Which is heavier?

EVALUATION

1. Is a "Square" 2D or 3D?
2. Is a "Cube" 2D or 3D?
3. What unit do we use to measure water? (ml or L?)
4. What unit do we use to measure a bag of rice? (g or kg?)
5. Does a triangle have corners? How many?
6. Which tool measures weight? (Ruler or Scale?)

ASSIGNMENT

1. Draw one 2D shape and one 3D shape. Label them.
 2. Write "Litre" or "Kilogram" for these items: a) A bottle of Coke. b) A bag of onions.
 3. Count how many corners a rectangle has.
-

WEEK 10: APPLIED PROBLEM SOLVING (ALL CONCEPTS).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Solve mixed problems involving money, time, and measurement.
- Apply addition and subtraction to real-life scenarios.
- Solve simple quantitative reasoning problems.

CONTENT



Mixed Operations: We use everything we learned in Primary 2 in one big problem.

- **Money + Time:** "If a bus ride costs ₦50 and takes 1 hour, and I have ₦100, can I pay for a return trip (go and come back)?"
- Cost for return: $\text{₦}50 + \text{₦}50 = \text{₦}100$.
- I have exactly ₦100. **Yes, I can pay.**

2. Real-Life Scenarios: Let's imagine a scenario:

- Mummy wants to bake a cake.
- She needs **500g** of flour (Weight).
- She needs **200ml** of milk (Capacity).
- She puts it in a **Square** baking tin (Shape).
- She bakes it for **1 hour** (Time).

- This shows how we use different maths in one activity.

3. Quantitative Reasoning: These are puzzle problems that make us think logically.

- **Logic Puzzle 1:**

- If Triangle = 2 and Square = 4.
- What is Triangle + Square? $2 + 4 = 6$.

- **Shape Patterns:**

- Look: Circle, Square, Circle, Square, Circle, ____
- The pattern repeats. What comes next? **Square**.

EVALUATION

1. Solve: $\text{R}20 + \text{R}10 - \text{R}5$. (Do addition first, then subtraction).
2. If I have a litre (1 L) of water and I drink half, is the bottle empty? (No, it is half-full).
3. Complete the pattern: Triangle, Triangle, Circle, Triangle, Triangle, ____.
4. How many minutes are in 1 hour? (Revision: 60 minutes).
5. Add $3\text{kg} + 2\text{kg}$.

ASSIGNMENT

1. Create a simple shopping list with prices for 3 items (e.g., Biscuit $\text{R}10$, Sweet $\text{R}5$, Pencil $\text{R}15$). Add the total cost.
 2. Draw a repeating pattern using Squares and Circles (e.g., $\square \circ \square \circ \square \circ$).
 3. If $A = 5$ and $B = 2$, find $A - B$.
-

WEEK 11: REVISION.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Recall definitions of weight, capacity, and area.
- Identify 2D and 3D shapes correctly.
- Interpret simple data from a chart.
- Attempt sample examination questions confidently.

CONTENT

Comprehensive Review: We go through the topics of the term one by one.

- **Week 1-3:** Measurement (Weight, Capacity, Area).
- **Week 4 & 7:** Shapes (2D and 3D).
- **Week 8:** Data (Tally marks and Pictograms).

2. Mock Exam Drill: Teacher writes sample questions on the board similar to the final exam. Pupils answer in their exercise books. Teacher goes through the answers immediately to correct mistakes.

EVALUATION

1. What is the difference between a square and a cube? (One is flat, the other is solid).
2. 5 tally marks look like what? (|||||).
3. Which is heavier: 1kg of stones or 1g of feather? (1kg is much heavier than 1g).
4. How many sides does a triangle have?
5. Name a container used to measure capacity. (Measuring cup or jug).

ASSIGNMENT

1. Study all notes from Week 1 to Week 10.
2. Practice drawing all the shapes (2D and 3D).
3. Prepare for the end-of-term examination.